

NITHYASRI SRIKUMAR

nithsri@seas.upenn.edu • github.com/nithyasri0913 • linkedin.com/nithyasri-s/

EDUCATION

University of Pennsylvania, School of Engineering and Applied Science
Master of Science in Engineering: Computer and Information Science

Philadelphia, PA
May 2027

SRM Institute of Science and Technology, College of Engineering
Bachelor of Technology in Computer Science and Engineering
Cumulative GPA: 3.98/4.00

Chennai, India
May 2027

SKILLS

Languages: Python, C/C++, Java, JavaScript, R, SQL

Frameworks and databases: React, Spring Boot, Express, Flask, PostgreSQL, MongoDB, Firebase, DB2

Libraries: Pandas, NumPy, Matplotlib, OpenCV, Scikit-Learn, Librosa, TensorFlow, PyTorch, Hugging Face Transformers, Tokenizers, NLTK

Concepts: Machine Learning, NLP, Computer Vision, Deep Learning, Data Structures & Algorithms

PROFESSIONAL EXPERIENCE

Wells Fargo, Software Engineer, Aug 2023 – Jul 2025

Bengaluru, India

- Modernized legacy mainframe systems into cloud-ready web apps, decommissioning three monolithic services and reducing infrastructure costs by 20%, through React.js (Redux, hooks) front end and Spring Boot microservices.
- Improved client-side performance by 30%, validated via Lighthouse and Core Web Vitals, by reducing React bundle size 40%, implementing code-splitting, and optimizing state management.
- Enhanced backend stability and maintainability, lowering production incidents by 25% over a year, by re-engineering business logic into modular Java services, upgrading React (16→18) and Java (11→17), and improving code quality via SonarQube.
- Implemented secure enterprise authentication with SSO, reducing auth-related defects by 40%, by integrating Ping Identity, OAuth2, JWT, and role-based access control (RBAC).

Wells Fargo, Summer Intern, Jun 2022 – Aug 2022

Virtual

- Built a stock market charting web app, evidenced by production deployment and daily active users, using React.js, Node.js, Firebase, and custom APIs as a key differentiator.
- Reduced data latency and optimized server load, validated via real-time benchmarks, by implementing dynamic polling intervals and real-time data fetching with Node.js and Firebase.
- Enhanced user engagement and insights, measured by increased watchlist interactions and chart views, by developing personalized watchlists and interactive market trend visualizations using Chart.js.
- Delivered predictive investment insights, demonstrated through successful stock price forecasts in production, by integrating LSTM models using Python (Keras/TensorFlow) and visualizing predictions on the React front end.

PROJECTS

Style Transfer using CycleGANs, *Team of 2*, Sep 2022 – Dec 2022

- Achieved high-quality artistic style transfer on images, measured by visually enhanced outputs, by developing and optimizing CycleGAN generator/discriminator networks with perceptual loss and instance normalization.

Collusion Detection in Auctions Using Clustering & Statistical Analysis, Jan 2022 – May 2022

- Detected dynamic collusion and enhanced auction fairness by identifying anomalous bidder groups and bid rotations using DBSCAN clustering, Markov Chains for sequence modeling, and Z-Score statistical analysis.

Image Captioning using ResNet-50 and LSTM with Attention Mechanism, Sep 2021 – Dec 2021

- Generated descriptive captions for 10k+ images, measured by BLEU and CIDEr scores, by extracting features with ResNet-50 and training an attention-enhanced LSTM.
- Enhanced caption relevance and alignment with visual regions, through hyperparameter tuning and attention visualization, demonstrating ability to handle large-scale, complex vision-language tasks.

RESEARCH

Multimodal Emotion Recognition from Audio and Video using CNN, Apr 2024

Chennai, India

- Achieved multimodal emotion recognition from audio and video, improving accuracy by over 60% compared to unimodal models, by combining audio features (STFT, MFCC via Librosa) and video features (facial landmarks with OpenCV) and applying CNNs with decision-level fusion.
- Demonstrated robust emotion classification, through a live application with real-world use in conversational agents, recommendation systems, and smart environments, validated through precision, recall, and F1-score, by integrating CNN-based facial expression models and systematically evaluating multimodal predictions.

ACHIEVEMENTS

Performance Based Scholarship, May 2020

Chennai, India

- Performance-based scholarship of Rs. 50,000 for securing the first rank (10 GPA) in the university examinations conducted in Year I.

CERTIFICATIONS AND TRAINING

- Machine Learning — Stanford Online (Coursera)
- Fundamentals of Deep Learning — NVIDIA
- Natural Language Processing with Probabilistic Models — DeepLearning.AI
- Build Generative Adversarial Networks (GANs) — DeepLearning.AI